

Writing results and limitations

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Writing the methods section

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Negative results and claim revision

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Running the primary experiment

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Writing results and limitations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Writing results and limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Candidate 10/15, baseline 9/15; no timings measured. Write result and two limits.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

B passes 9/12, A 8/12; three B failures all concern conflicting sources. No user study occurred. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A limitation should directly bound the corresponding claim rather than contradict an oversized headline. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Candidate10/15, baseline9/15; no timings measured. Write result and two limits.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled writing results and limitations evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of writing results and limitations. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Writing results and limitations

Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection.

A supports 9/12 answers; B supports 10/12. No user learning outcomes are recorded.

Supported answers in one collection

supported answers out of 12

A



B



Bar length shows the supplied value; the marked baseline is zero.

Writing results and limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Report the observation

A is 75%; B is about 83.3%, a one-case improvement.



2. Link to the method

Name the collection, rubric and twelve-case denominator.



3. State limits

Do not infer that children learn more from this answer-support measure.

CHANGE ONE THING

Write "B makes every student smarter".

Explain the changed result using the matching lesson.

Compute the supported-answer rates for 9/12 and 10/12. What actual outcome was measured, and why is a claim that every student becomes smarter outside this evidence?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Do not infer that children learn more from this answer-support measure.

1. Report the observation
2. Link to the method
3. State limits

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: State limits Do not infer that children learn more from this answer-support measure. Results describe what was measured; limitations prevent substitution of a different outcome or population. A small descriptive improvement needs its scope attached.

Practice 2: explain the deciding evidence

Changed-case practice: Write "B makes every student smarter". Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Remove that sentence as unsupported by the study.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Remove that sentence as unsupported by the study. The measured unit was answers, not student understanding or intelligence.

Repair a missing building block

Review Writing the methods section (ai-7/30). Goal: Explain the mechanism and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Review Negative results and claim revision (ai-7/28). Goal: Explain the mechanism and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the prediction that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the mechanism and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

